MADHUBALA V

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CAREER SUMMARY

Aspiring Data Science graduate with skills in Python, SQL, Tableau, and data analytics. Skilled in data visualization, exploratory data analysis, predictive modeling, and machine learning through academic and personal projects. Passionate about transforming data into actionable insights and practical solutions.

EDUCATION

M.Sc. Applied Data Science | SRM Institute of Science and Technology | **CGPA: 9.54** **2024 - 2026**

B.Sc. Data Science | Sri Ramachandra Faculty of Engineering and Technology | **CGPA: 8.05** **2020 – 2023**

INTERNSHIP

Data Science Intern | L&T Construction **April 2022 – June 2022**

- **Waste Bin Analytics:** Analyzed 11K+ sensor records using data cleaning, EDA, visualizations, and Linear Regression; refined fill-level predictions, achieving 94–98% reported accuracy.
- **Mining Analytics:** Analyzed 737K+ records using data preprocessing, EDA, visualization, and regression modelling to predict silica impurity levels, achieving 67% accuracy.
- **Loan Analytics:** Analyzed 40K+ applicant records using data cleaning, EDA, and Multiple Linear Regression to predict loan amounts; evaluated model performance with a 39% reported score.

TECHNICAL SKILLS

Languages: Python | SQL | HTML | CSS
Data Analysis: EDA | Statistical Analysis | Predictive Analytics
Visualization: Tableau | Excel | Matplotlib | Seaborn
Machine Learning: Scikit-learn | TensorFlow | Keras
Tools: Jupyter Notebook | Google Colab
Gen AI Tools: ChatGPT | Google Gemini | DeepSeek | Canva

CERTIFICATIONS

- **GenAI Powered Data Analytics Job Simulation** | Tata Group (Forage) | 2026
- **Deloitte Data Analytics Job Simulation** | Deloitte (Forage) | 2026
- **Statistics for Data Science |** Great Learning **|** 2024
- **SQL and Relational Database |** Cognitive Class (IBM) | 2023

ACADEMIC PROJECTS

Knee Osteoarthritis Detection

Python | TensorFlow | Streamlit| Transfer Learning

- Analyzed knee X-rays across 5 KL grades (0–4) using OpenCV and Pandas; performed image preprocessing and augmentation.
- Trained and compared ResNet50, EfficientNetB3 and DenseNet121, achieving 69.32% test accuracy with ResNet50.
- Developed a Streamlit app for new X-ray predictions, displaying KL grades, confidence scores, class probabilities and Grad-CAM heatmaps.

Flipkart Sales & Customer Insights Dashboard

Tableau | Data Visualization | Python

- Cleaned and analyzed 20K+ product records using Python and Pandas.
- Built an interactive Tableau dashboard; identified Clothing (6,341) as the largest category and 40.5% average discount.
- Recommended targeted promotions and improved Flipkart Advantage coverage (3.92% of listings).

Heart Attack Prediction

Python | Gradio | Scikit – learn | XGBoost

- Analyzed 1,319 patient records using Pandas, NumPy, Matplotlib and Seaborn; performed EDA and feature engineering.
- Compared 7 ML models; achieved 98.1% accuracy, 98.8% recall and 98.5% F1-score with XGBoost.
- Identified key risk factors (Troponin, CK-MB) and built a Gradio interface for patient risk prediction using 3 ML models.